

AI engineering fundamentals

A one-sitting sheet. The long form is the concept cards. The PDF copy is [ai-engineering-cheatsheet.pdf](#).

Language model

A transformer estimates the next token. Softmax turns logits into probabilities. Greedy decoding takes the most likely token. Sampling draws one after temperature, top-k, or top-p. Lower temperature sharpens the distribution. Instruction tuning and preference tuning (RLHF, DPO) update weights. A prompt does not. The context window is a token budget, not the parameter count.

Tokens

Count with the tokenizer that belongs to the model. BPE merges frequent pairs (Sennrich, Haddow, and Birch, 2016). WordPiece is tied to BERT. SentencePiece can skip a separate word splitter. Byte-level BPE consumes UTF-8 (RFC 3629). Special tokens count. Words, characters, and tokens are three different numbers.

Embeddings

Same model, same preprocessing, same metric. On unit-length vectors, cosine similarity and dot product rank alike. A bi-encoder stores document vectors ahead of time. A cross-encoder reranks a short list. HNSW and inverted-file indexes are approximate. Product quantization compresses vectors. Equal dimension does not make two models interchangeable.

RAG, eight steps

1. **Ingest.** Accept the document, its source, its date, and who may read it.
2. **Chunk.** Split the document into passages you can return on their own.
3. **Embed.** Map each chunk with a single embedding model.
4. **Index.** Store the vector, the passage, and the filter fields.
5. **Retrieve.** Embed the query the same way and take the top k.
6. **Rerank.** Rescore the short list, often with a cross-encoder.
7. **Generate.** Answer from the question and the passages you kept.
8. **Cite.** Point each material claim at a retrieved passage.

Hybrid search runs BM25 beside the vectors. If the passages do not contain the answer, say so. A citation is a pointer you can still reject.

Agent loop

1. **Plan.** Name the goal, the constraint, and what done means.
2. **Act with tools.** The model proposes a call. The host checks it, then runs it.
3. **Observe.** Record the result, including errors and empty answers.
4. **Reflect.** Compare the result with the goal. Keep the plan, revise it, or stop.

Stop when the goal is met, the budget is spent, the same error repeats, or a person must approve. The return path is “not done,” and it is still inside the budget. The host runs tools.

Tools and MCP

A tool schema names the operation and the JSON shape of its arguments. The model returns a request. The host validates and runs it, or refuses. MCP carries that relationship in JSON-RPC 2.0, which is its own specification and not an IETF RFC. A host uses a client to reach a server. Servers expose tools, readable resources, and prompt templates. Connecting a server is a trust decision. A sentence inside a tool result does not grant a new tool.

Evaluation

- Recall@k: fraction of known relevant items that appear in the top k.
- Precision@k: fraction of the k hits that are relevant.
- MRR: average of $1 / \text{rank}$ of the first relevant hit.
- nDCG: rewards relevant items placed earlier.
- Grade the answer separately: groundedness, exact match, or a written rubric.
- Freeze a golden set. Hold some items out of prompt tinkering.
- Calibrate an LLM judge against people. Record the judge's model.
- Report latency, cost, refusals, and the previous version's score.

Seven failures

1. A word count used as a token budget.
2. Two embedding models mixed because the dimension matched.
3. A citation whose passage does not say the claim.
4. A long context treated as proof the middle was used.
5. An agent loop with no step or cost cap.
6. A tool result treated as permission to call a different tool.
7. A score reported on the items you edited the prompt to pass.